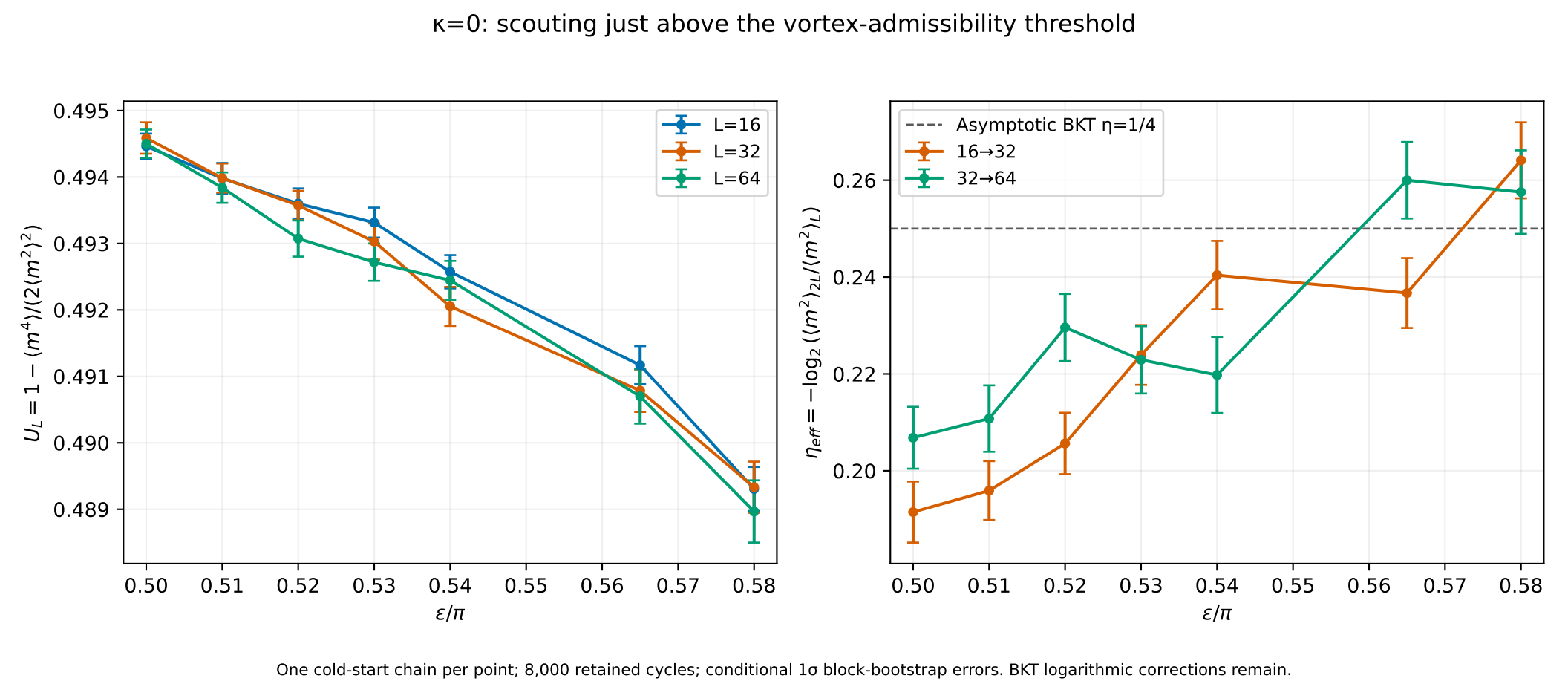
κ=0: scouting just above the vortex-admissibility threshold
$U_L = 1 - \langle m^4 \rangle / (2\langle m^2 \rangle^2)$
ε/π
L=16
L=32
L=64
$\eta_{eff} = -\log_2 (\langle m^2 \rangle_{2L} / \langle m^2 \rangle_L)$
Asymptotic BKT η=1/4
16→32
32→64
One cold-start chain per point; 8,000 retained cycles; conditional 1σ block-bootstrap errors. BKT logarithmic corrections remain.